

Quaternion-Based Robust Attitude Control for Uncertain Robotic Quadrotors

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Abstract—A robust nonlinear attitude control method is proposed for uncertain robotic quadrotors. The proposed controller is developed based on a nonlinear model with the quaternion representation and subject to parameter uncertainties, nonlinearities, and external disturbances. A new state feedback controller is proposed to restrain the effects of nonlinearities and uncertainties on the closed-loop control system. These uncertainties are considered as input equivalent disturbances and their effects are guaranteed to be attenuated. Experimental results are given to show good steady-state and dynamic tracking performance of the closed-loop system by the proposed robust control method compared with other nonlinear control methods.

Index Terms—Attitude control, quadrotor, quaternion, robust control, uncertainty.

I. INTRODUCTION

IN THE last decade, developments of unmanned helicopter systems have made significant progress (see [1]–[9] to mention a few). Applications of these aerial systems have been extended to some new fields including disaster assistance in industrial environment, urban search and rescue, and image acquisition in inaccessible areas [11]. Particular interests of the research community have been attracted to the quadrotors, because of their two main advantages over conventional helicopters [12].

- 1) Complex mechanical control linkages are not required for the rotor actuation.
- 2) For a given airframe size, the application of four rotors guarantees that each individual rotor of quadrotors is smaller than the main rotor of conventional helicopters.

Therefore, many efforts have been made to control the quadrotors in the control circle. Based on accurate mathematical model, some control approaches have been proposed to achieve the automatic flight of the vehicles (see, for instance, [11]–[17]).

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However, the quadrotor dynamics involves multiple uncertainties including parameter perturbations, coupling, nonlinearity, and external disturbances. First, the payload configuration may be changed due to different mission requirements, which would result in the variations of the quadrotor weight and inertia tensor. On the other hand, certain vehicle parameters, which depend on air density, vehicle rotational velocities, and aerodynamic characteristics, are usually regarded as constants in the vehicle modeling. Therefore, the vehicle dynamics is subject to parameter perturbations. Second, there exist serious nonlinearity and inter-axis coupling between the attitude angles and their rates of change in the quadrotor model. Furthermore, the vehicle dynamics is prone to being influenced by sudden or persistent wind gusts. The external disturbances from wind gusts may seriously affect the vehicle flight performance and even lead to the instability of the closed-loop system. Therefore, new robust strategies have been developed to deal with the uncertainty rejection problems. In [18] and [19], new developed sliding mode control approaches were proposed to achieve disturbance rejection during the flight of quadrotors. But the dynamical tracking performance of the closed-loop control systems cannot be guaranteed by the discontinuous control methods. In [20] and [21], adaptive control algorithms were discussed to achieve the robust trajectory tracking control for quadrotors. Based on mixed H_2 and H_∞ performance cost functions, a suboptimal controller design method was presented in [22]. But, the influence of nonlinearity and coupling on the closed-loop control systems cannot be restrained by the standard H_∞ control approach in the whole frequency range. Furthermore, most of the previous works focused on designing robust controllers based on the Euler angle-based representation to describe the rotational dynamics of the helicopters (see, e.g., [1]–[22]). However, the representation suffers from singularity problems [23].

Unit quaternion representations can enable the motion control systems to be all-attitude capable [24]. Furthermore, the quaternion-based attitude propagation algorithms are globally nonsingular and numerically robust and thereby have attracted much attention in the control and attitude estimation community [23]–[28]. But these representations are nonminimal and constrained to have unit length, which may lead to difficulties when attempting to design robust flight controllers. In [25], a nested saturation control method was proposed based on rotational motion equations described by the quaternion representation. In [23], the quaternion representation was applied to avoid singularities inherent to the rotation matrix defined by Euler angles and the exponential attitude stabilization was achieved by compensation of the Coriolis and gyroscopic

torques. However, the effects of uncertainties were not fully discussed in the performance analysis of the designed closed-loop control systems as shown in [23] and [25].

Many of the previous works focused on improve the robustness properties of the helicopter systems based on the quaternion representations by the adaptive control approaches. Direct adaptive control approach based on the neural network was proposed in [24] to achieve trajectory tracking for autonomous helicopters. In [26], an adaptive controller was designed to achieve robust attitude control of uncertain helicopters described by the quaternions-based nonlinear equations. In [27], a novel nonlinear adaptive output regulator was designed to control the vertical motion while stabilizing the lateral and horizontal positions of a helicopter. But the dynamical tracking performance of the adaptive closed-loop systems cannot be specified and the effects of time-varying disturbances on the control systems were not fully discussed in their stability analysis. Actually, the definitions of the addition and subtraction of the quaternions are nonlinear and complex, which might lead to the difficulties on evaluating the tracking performance of the closed-loop control system against the influence of uncertainties.

This paper aims at addressing the quaternion-based robust attitude control problem of quadrotors, whose dynamics involves parameter perturbations, coupling, nonlinearity, and external disturbances. A nominal model is first obtained by ignoring the coupling terms between the angular velocities, while the coupling terms, parametric uncertainties, and external uncertainties are considered as the input equivalent disturbances. The proposed controller consists of two parts: 1) a nominal controller; and 2) a robust compensator. The nominal controller is designed to achieve the desired tracking for the nominal system, while the robust compensator is introduced to attenuate the influence of the input equivalent disturbances.

Compared to previous studies on the robust control problem for the quadrotor dynamics described by the quaternions, the effects of various uncertainties on the closed-loop control system are guaranteed to be attenuated. The dynamical tracking performance of the proposed control system can be specified and the tracking errors of the translational positions are proven to converge into a given neighbourhood of the origin. Furthermore, experimental results of tracking missions for sine wave references under persistent wind gusts are given to show the advantages of the proposed control method. Compared to previously published paper [29], this paper develops a first-order robust compensator with comparatively lower controller gain and a simple structure to restrain the effects of uncertainties. Furthermore, quaternion-based attitude representations have been applied to avoid the singularity problem here. Moreover, lower limits of the robust compensator parameters are given and attitude tracking errors can enter a given neighborhood of the origin in a given finite time. Besides, the quadrotor tracking missions are more challenging and the tracking performance is improved in this paper.

This paper is organized as follows. In Section II, the quaternion-based mathematical model is introduced. In Section III, a new controller design method is presented and

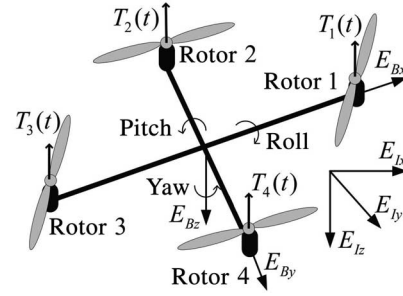


Fig. 1. Schematic of the quadrotor.

the tracking performance of the proposed closed-loop control system is analyzed. Experimental results are illustrated and discussed in Section IV. Conclusion remarks are drawn in Section V.

II. NONLINEAR MODEL FORMULATION

As depicted in Fig. 1, the quadrotor includes a rigid cross frame and four rotors. The longitudinal pair of rotors (1, 3) rotates counterclockwise and the lateral pair of rotors (2, 4) rotates clockwise. If a pitching motion is desired, one can increase the thrust $T_1(t)$ produced by Rotor 1 and decrease $T_3(t)$. On the other hand, a rolling motion can be achieved by changing $T_2(t)$ and $T_4(t)$ in a similar way. Reducing the thrust of one rotor pair while increasing the thrust of the other rotor set, a yawing motion can be obtained due to the induced reactive torque produced by the four rotors. Let $E_I = \{E_{Ix}, E_{Iy}, E_{Iz}\}$ denote an earth fixed inertial frame and $E_B = \{E_{Bx}, E_{By}, E_{Bz}\}$ a body fixed frame rigidly attached to the vehicle with origin in the mass center, as shown in Fig. 1. Then, as shown in [27], the dynamical model describing the rotation motion can be given by

$$\begin{cases} \dot{R}(t) = R(t)S(\omega_b(t)) \\ J\dot{\omega}_b(t) = -S(\omega_b(t))J\omega_b(t) + \tau_b(t) \end{cases} \quad (1)$$

where $\omega_b(t) = [\omega_{bi}(t)]_{3 \times 1}$ is the angular velocity in the frame E_B , J the inertia tensor of the body, which is a symmetric and positive definite constant matrix, $\tau_b(t) = [\tau_{bi}(t)]_{3 \times 1}$ the external torque acting on vehicle in the frame E_B , $R(t) \in SO(3)$ the rotational matrix mapping vectors expressed in E_B into the vectors expressed in E_I , and

$$S(\omega_b(t)) = \begin{bmatrix} 0 & -\omega_{b3}(t) & \omega_{b2}(t) \\ \omega_{b3}(t) & 0 & -\omega_{b1}(t) \\ -\omega_{b2}(t) & \omega_{b1}(t) & 0 \end{bmatrix}.$$

In this paper, unit quaternion representations are used to describe the rotational motion to avoid singularity problems that Euler angle-based representations have. Let $q(t) = [q_1(t) \bar{q}_1^T(t)]^T \in \mathcal{R}^{4 \times 1}$ denote unit quaternions, where $q_1(t)$ and $\bar{q}_1(t) = [q_2(t) q_3(t) q_4(t)]^T \in \mathcal{R}^{3 \times 1}$ are the scalar part and the vector part of the quaternion, respectively, and satisfy that

$$q_1^2(t) + \|\bar{q}_1(t)\|_2^2 = 1.$$

From [27], the rotation matrix can be given by four quaternions as shown at the bottom of the page

Actually, the first equation in (1) can be replaced by the following quaternion propagation equations as shown in [27]

$$\begin{cases} \dot{q}_1(t) = -\frac{1}{2}\bar{q}_1^T(t)\omega_b(t) \\ \dot{\bar{q}}_1(t) = \frac{1}{2}[q_1(t)I_3 + S(\bar{q}_1(t))]\omega_b(t) \end{cases} \quad (2)$$

where I_n is an $n \times n$ identity matrix. The external torques in the frame E_B can be obtained by the following equations:

$$\begin{cases} \tau_{b1}(t) = l_d(T_2(t) - T_4(t)) \\ \tau_{b2}(t) = l_d(T_1(t) - T_3(t)) \\ \tau_{b3}(t) = \tau_1(t) - \tau_2(t) + \tau_3(t) - \tau_4(t) \end{cases} \quad (3)$$

where $l_d > 0$ denotes the distance from each rotor to the mass center of the vehicle and $\tau_i(t)$ ($i = 1, 2, 3, 4$) are the reactive torques generated by the four rotors respectively. The thrusts and reactive torques produced by each rotor can be given by

$$\begin{cases} T_i(t) = k_{T\omega}\omega_{ri}^2(t) \\ \tau_i(t) = k_{\tau\omega}\omega_{ri}^2(t), \quad i = 1, 2, 3, 4 \end{cases} \quad (4)$$

where $\omega_{ri}(t)$ ($i = 1, 2, 3, 4$) are the rotational speeds of Rotor i respectively, and $k_{T\omega}$ and $k_{\tau\omega}$ are positive constants relative to the air density, the drag constant, and geometrical features of the blades such as chord length, pitch angle, and blade shape. Let $u_g = \omega_{r1}^2(t) + \omega_{r2}^2(t) + \omega_{r3}^2(t) + \omega_{r4}^2(t)$, which is set to be a positive constant to produce thrusts to approximately counteract the effects of gravity in hovering. Design the control inputs $u_i(t)$ ($i = 1, 2, 3$) as

$$\begin{cases} u_1(t) = \omega_{r2}^2(t) - \omega_{r4}^2(t) \\ u_2(t) = \omega_{r1}^2(t) - \omega_{r3}^2(t) \\ u_3(t) = \omega_{r1}^2(t) - \omega_{r2}^2(t) + \omega_{r3}^2(t) - \omega_{r4}^2(t). \end{cases} \quad (5)$$

There already exists a power distribution board to distribute the four control inputs $u_i(t)$ ($i = 1, 2, 3$) and u_g to the four rotors. Actually, there exists inner dynamics between each control input and the corresponding rotational speed of the rotor. The convergence of real rotational velocities to the desired ones is much faster than that of the attitude tracking errors to zeros, and thereby the inner dynamics is ignored here.

In this paper, the desired attitude references are denoted by $q^r(t) = [q_1^r(t) \ q_2^r(t) \ q_3^r(t) \ q_4^r(t)]^T \in \mathcal{R}^{4 \times 1}$. The desired references and their first and second derivatives are assumed to be piecewise uniformly bounded. The attitude tracking error vector can be obtained by a nonlinear function $\tilde{Q}(q(t), q^r(t)) \in \mathcal{R}^{3 \times 1}$. As shown in [24], an expression for $\tilde{Q}(q(t), q^r(t))$ can be given by

$$\begin{aligned} \tilde{Q}(q(t), q^r(t)) &= 2\text{sgn}\left(\sum_{i=1}^4 q_i^r(t)q_i(t)\right) \\ &\times \begin{bmatrix} -q_1^r(t)q_2(t) + q_2^r(t)q_1(t) + q_3^r(t)q_4(t) - q_4^r(t)q_3(t) \\ -q_1^r(t)q_3(t) - q_2^r(t)q_4(t) + q_3^r(t)q_1(t) + q_4^r(t)q_2(t) \\ -q_1^r(t)q_4(t) + q_2^r(t)q_3(t) - q_3^r(t)q_2(t) + q_4^r(t)q_1(t) \end{bmatrix}. \end{aligned}$$

Define error vector $e(t)$ as $e(t) = [e_q^T(t) \ e_\omega^T(t)]^T \in \mathcal{R}^{6 \times 1}$, where $e_q(t) = [e_{qi}(t)]_{3 \times 1} = \tilde{Q}(q(t), q^r(t))$ and $e_\omega(t) = [e_{\omega i}(t)]_{3 \times 1} = \omega_b(t) - \omega_b^r(t)$ with $\omega_b^r(t)$ the desired rotational speeds in the frame E_B . Actually, $e_q(t)$ is the attitude tracking error vector and $e_\omega(t)$ is rotational velocity tracking error vector. From [24], the tracking error dynamics can be obtained by directly differentiating $e(t)$ as

$$\dot{e}(t) = \begin{bmatrix} \omega_b(t) - \omega_b^r(t) \\ \dot{\omega}_b(t) - \dot{\omega}_b^r(t) \end{bmatrix}.$$

Define

$$\begin{aligned} u(t) &= [u_1(t)u_2(t)u_3(t)]^T \in \mathcal{R}^{3 \times 1}, \\ B_\tau &= \text{diag}(k_{T\omega}l_d, k_{T\omega}l_d, k_{\tau\omega}). \end{aligned}$$

Combining (1) and (3)–(5), one can obtain that

$$\dot{\omega}_b(t) = -J^{-1}S(\omega_b(t))J\omega_b(t) + J^{-1}B_\tau u(t).$$

Let $B_\omega = J^{-1}B_\tau$. Then, it follows that

$$\begin{cases} \dot{e}_q(t) = e_\omega(t), \\ \dot{e}_\omega(t) = -J^{-1}S(\omega_b(t))J\omega_b(t) + B_\omega u(t) - \dot{\omega}_b^r(t) + d_{e\tau}(t) \end{cases} \quad (6)$$

where $d_{e\tau}(t) \in \mathcal{R}^{3 \times 1}$ is external bounded uncertainty, e.g., wind disturbance acting on the vehicle. The quadrotor parameter matrix B_ω can be divided into two parts as

$$B_\omega = B_{\omega N} + B_{\omega \Delta}$$

where the subscript N denotes the nominal parameter matrix and Δ denotes the parameter uncertainties. The matrix $B_{\omega N}$ is a symmetric and positive definite matrix and is assumed to satisfy that $\|B_{\omega N}^{-1}B_{\omega \Delta}\|_1 < 1$. Define

$$A = \begin{bmatrix} O_3 & I_3 \\ O_3 & O_3 \end{bmatrix}, \quad B = \begin{bmatrix} O_3 \\ B_{\omega N} \end{bmatrix}$$

where O_3 are a 3×3 zero matrix. Then, one can rewrite the error model (6) as follows:

$$\dot{e}(t) = Ae(t) + B(u(t) + \Delta(t)) \quad (7)$$

where $\Delta(t) \in \mathcal{R}^{3 \times 1}$ is the input equivalent disturbances and has the following form

$$\begin{aligned} \Delta(t) &= B_{\omega N}^{-1}(-J^{-1}S(\omega_b(t))J\omega_b(t) + B_{\omega \Delta}u(t) \\ &\quad - \dot{\omega}_b^r(t) + d_{e\tau}(t)). \end{aligned} \quad (8)$$

In this paper, the following problem of designing a robust and nonlinear attitude controller will be investigated: for a given positive constant ε , a given initial state $e(0)$, and a positive

$$R(t) = \begin{bmatrix} 1 - 2q_3^2(t) - 2q_4^2(t) & 2q_2(t)q_3(t) - 2q_1(t)q_4(t) & 2q_2(t)q_4(t) + 2q_1(t)q_3(t) \\ 2q_2(t)q_3(t) + 2q_1(t)q_4(t) & 1 - 2q_2^2(t) - 2q_4^2(t) & 2q_3(t)q_4(t) - 2q_1(t)q_2(t) \\ 2q_2(t)q_4(t) - 2q_1(t)q_3(t) & 2q_3(t)q_4(t) + 2q_1(t)q_2(t) & 1 - 2q_2^2(t) - 2q_3^2(t) \end{bmatrix}.$$

constant T^* , the attitude tracking errors and rotational velocity tracking errors are bounded and satisfy that the inequality: $\max_k |e_k(t)| \leq \varepsilon \forall t \geq T^*$.

III. ROBUST ATTITUDE CONTROLLER DESIGN

A systematic robust controller design procedure is presented in this section for the quadrotor model based on the quaternion representation. The designed control input $u(t)$ contains two terms: 1) the nominal control input $u^N(t) = [u_i^N(t)]_{3 \times 1}$; and 2) robust compensating input $u^R(t) = [u_i^R(t)]_{3 \times 1}$, and satisfies that

$$u(t) = u^N(t) + u^R(t). \quad (9)$$

The nominal control input is introduced to achieve the desired tracking for the nominal closed-loop system and robust compensating input is added to restrain the influence of the input equivalent disturbances $\Delta(t)$.

First, the nominal controller can be constructed as follows (see, [24])

$$u^N(t) = -Ke(t) = -[K_q \ K_\omega] e(t) \quad (10)$$

where the linear gain matrix $K = [k_{ij}]_{3 \times 6}$ is required to be chosen such that $A_K = A - BK$ is Hurwitz. Combining (7), (9), and (10), one can obtain that

$$\dot{e}(t) = A_K e(t) + B(u^R(t) + \Delta(t)). \quad (11)$$

Actually, the error model (11) can be considered as the nominal closed-loop control system by ignoring the input equivalent disturbances $\Delta(t)$. By choosing the linear gain matrix K , one can set the nominal closed-loop system with desired poles.

Then, the robust compensator is introduced to restrain the effects of uncertainties $\Delta(t)$. The desired robust compensating input $u^R(t)$ is given by $-\Delta(t)$. However, it can be seen from (8) that $\Delta(t)$ involves uncertain parameters and external uncertainties and thus the desired robust compensator cannot be realized directly. From (7), one can have that

$$\Delta(t) = B_{\omega N}^{-1} \dot{e}_\omega(t) - u(t). \quad (12)$$

However, $\dot{e}_\omega(t)$ in (12) are unmeasured variables. Therefore, the robust filters $F_i(s) (i = 1, 2, 3)$ are introduced as follows:

$$F_i(s) = \frac{f_i}{s + f_i}, \quad i = 1, 2, 3 \quad (13)$$

where s is the Laplace operator and $f_i (i = 1, 2, 3)$ are positive constants. Actually, $F_i(s) (i = 1, 2, 3)$ are low-order filters and have the following property: if $f_i (i = 1, 2, 3)$ are selected with larger positive values, the bandwidths of these filters would be wider and thereby the gains of the filters would get closer to 1 (see, [30]–[32] to mention a few). Therefore, construct the robust compensating input as follows:

$$u^R(s) = -F(s)\Delta(s) \quad (14)$$

where $F(s) = \text{diag}(F_1(s), F_2(s), F_3(s))$. Substituting (12) into (14), one can obtain that

$$u_i^R(s) = -\frac{s f_i}{s + f_i} c_{3,i}^T B_{\omega N}^{-1} e_\omega(s) + \frac{f_i}{s + f_i} u_i(s), \quad i = 1, 2, 3$$

where $c_{n,i}$ is an $n \times 1$ vector with a one on the i th row and zeros elsewhere. The robust compensator can be realized as follows:

$$\begin{cases} \dot{z}_i^R(t) = -f_i z_i^R(t) + f_i c_{3,i}^T B_{\omega N}^{-1} e_\omega(t) + u_i(t) \\ u_i^R(t) = -c_{3,i}^T B_{\omega N}^{-1} e_\omega(t) + f_i z_i^R(t), \quad i = 1, 2, 3. \end{cases} \quad (15)$$

The whole controller can be described by the following equations:

$$\begin{aligned} u_i(s) = & -k_i e_{q_i}(s) - s f_i c_{3,i}^T B_{\omega N}^{-1} e_\omega(s) / (s + f_i) \\ & - k_{i+3} e_{\omega_i}(s) + f_i u_i(s) / (s + f_i), \quad i = 1, 2, 3. \end{aligned} \quad (16)$$

The robustness properties of the closed-loop control system can be summarized by the following theorem.

Theorem 1: For the closed-loop control system consisting of the quadrotor plant (6), there exists a positive constant f_{LL}^* such that the robust attitude control problem described in Section II is solvable by the feedback control laws as shown in (10) and (15) with robust compensator parameters $f_i (i = 1, 2, 3)$ satisfying that $f_i \geq f_{LL}^*$.

Proof: From (11) and (14), one can obtain that

$$\dot{e}(t) = A_K e(t) + B(I_3 - F)\Delta(t). \quad (17)$$

It follows that

$$\|e\|_\infty \leq \pi_{e(0)} + \delta \|\Delta\|_\infty \quad (18)$$

and

$$\max_k |e_k(t)| \leq \max_k |c_{6,k}^T e^{A_K t} e(0)| + \delta \|\Delta\|_\infty \quad (19)$$

where

$$\pi_{e(0)} = \maxsup_{k \ t \geq 0} |c_{6,k}^T e^{A_K t} e(0)|, \delta = \|(sI_6 - A_K)B(I_3 - F)\|_1.$$

If the positive parameters $f_i (i = 1, 2, 3)$ are larger, the gains of the robust filter F would get closer to I_3 . In this case, δ could be made as small as desired. Actually, there exists a positive constant λ_f such that $\delta \leq \lambda_f / f_{\min}$, where $f_{\min} = \min_i f_i$. From (8), there exist positive constants π_{ue2} , π_{ue1} , and π_{uec} such that

$$\begin{aligned} \|\Delta\|_\infty \leq & \pi_{ue2} \|e_\omega\|_\infty^2 + \pi_{ue1} \|e_\omega\|_\infty \\ & + \|B_{\omega N}^{-1} B_{\omega \Delta}\|_1 \|u\|_\infty + \pi_{uec}. \end{aligned}$$

Let $\rho = \|B_{\omega N}^{-1} B_{\omega \Delta}\|_1 < 1$. Combining (9), (10), and (14), one can have that

$$\|u\|_\infty \leq \|K\|_1 \|e\|_\infty + \|\Delta\|_\infty.$$

Then, it follows that:

$$\|\Delta\|_\infty \leq \pi_{\Delta e2} \|e\|_\infty^2 + \pi_{\Delta e1} \|e\|_\infty + \pi_{\Delta c} \quad (20)$$

where $\pi_{\Delta e2} = \pi_{ue2} / (1 - \rho)$, $\pi_{\Delta e1} = (\rho \|K\|_1 + \pi_{ue1}) / (1 - \rho)$, and $\pi_{\Delta c} = \pi_{uec} / (1 - \rho)$. If δ is enough small and the following inequality holds:

$$(\sqrt{\delta} + \delta)^{-1} \geq \pi_{\Delta e1} + \pi_{\Delta e2} \|e\|_\infty \quad (21)$$

then from (18) and (20), one obtains that

$$\|\Delta\|_{\infty} \leq \sqrt{\delta}^{-1}\pi_{e(0)} + (1 + \sqrt{\delta})\pi_{\Delta c}. \quad (22)$$

Substituting (22) into (18), one has that

$$\|e\|_{\infty} \leq \pi_{e(0)} + \sqrt{\delta}\pi_{ec} \quad (23)$$

where π_{ec} is a positive constant satisfying $\pi_{ec} \geq \pi_{e(0)} + (\sqrt{\delta} + \delta)\pi_{\Delta c}$. From the above expression, it can be observed that the attitude tracking errors and rotational velocity tracking errors are bounded. Actually, the attractive region of $e(t)$ can be obtained from (21) as

$$\left\{ e(t) : \|e\|_{\infty} \leq \pi_{\Delta e2}^{-1}(\sqrt{\delta} + \delta)^{-1} - \pi_{\Delta e2}^{-1}\pi_{\Delta e1} \right\}. \quad (24)$$

If $f_{\min}$ satisfies that $f_{\min} \geq f_{\min 1}, f_{\min 2}, f_{\min 3}$ where

$$\begin{cases} f_{\min 1} = 4\lambda(\pi_{\Delta e2}\|e(0)\|_{\infty} + \pi_{\Delta e1})^2 \\ f_{\min 2} = 4\lambda(\pi_{\Delta e2}\pi_{e(0)} + \pi_{\Delta e1})^2 \\ f_{\min 3} = \lambda \end{cases} \quad (25)$$

then $e(t)$ can remain inside the attractive region and (24) holds. In this case, from (19) and (22), one has that

$$\max_k |e_k(t)| \leq \max_k |c_{6,k}^T e^{A\kappa t} e(0)| + \sqrt{\delta}\pi_{ec}. \quad (26)$$

From the above inequality, it can be observed that the attitude tracking error $e(t)$ depends on the properties of the nominal closed-loop control system, and the term $\sqrt{\delta}\pi_{ec}$. By choosing K , one can set the rate of convergence of the nominal closed-loop control system as high as desired. Besides, by selecting the robust filter parameters $f_i (i = 1, 2, 3)$ with larger positive values, one can set δ as small as desired. So, for given positive constants ε and T^* and initial state $e(0)$, one can find a feedback matrix K and robust filter parameters $f_i (i = 1, 2, 3)$, such that $\max_k |c_{6,k}^T e^{A\kappa t} e(0)| \leq \varepsilon_N \forall t \geq T^*$ and $f_{\min} \geq f_{\min 1}, f_{\min 2}, f_{\min 3}, f_{\min 4}$, where $\varepsilon_N < \varepsilon$ is a positive constant, $\varepsilon_R = \varepsilon - \varepsilon_N$, and $f_{\min 4} = \lambda f_{ec}^2 / \varepsilon_R^2$. In this case, one can obtain that $\max_k |e_k(t)| \leq \varepsilon \forall t \geq T^*$.

Let $f_{LL}^* = \max \{f_{\min 1}, f_{\min 2}, f_{\min 3}, f_{\min 4}\}$, where f_{LL}^* represents the lower limit of the robust compensator parameters. It should be pointed out that f_{LL}^* determined by the above theoretical analysis can be conservative. Actually, in practical applications, the robust compensator parameters $f_i (i = 1, 2, 3)$ can be tuned online as follows: set $f_i (i = 1, 2, 3)$ with some initial values and run the whole closed-loop control system; if the tracking performance is unsatisfactory, set them with large values until the desired tracking performance can be obtained. Theoretically, if $f_i (i = 1, 2, 3)$ are larger, the bandwidths of the robust filters are wider and thus δ is smaller. In this case, from (19), one can see that the tracking error $e(t)$ would be smaller.

IV. EXPERIMENTAL RESULTS

The experimental quadrotor used in this section is a developed version of X-aircraft X650. Based on the mechanical frame, blades, and four motors of the toy helicopter, the

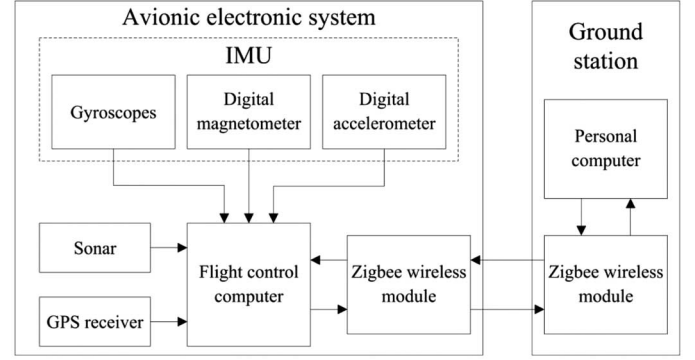


Fig. 2. Hardware configuration of quadrotor experimental platform.

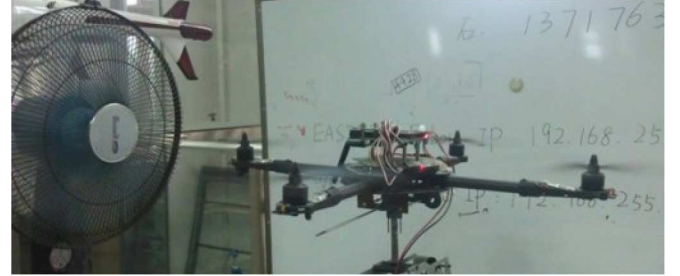


Fig. 3. Quadrotor experimental setup.

electronic circuitry and attitude sensors have been added as shown in Fig. 2. A digital compass, a digital accelerometer, and gyroscopes are applied to estimate the attitude information. An onboard DSP is used as the flight control computer to implement the proposed robust control algorithm with a sampling time of 2 ms. Since the attitude control problem is discussed in this paper, a stationary joint base is developed based on the swashplate of a conventional helicopter to restrict the three positions of the vehicle as shown in Fig. 3. The nominal parameter matrix is $B_{\omega N} = \text{diag}(10, 10, 15)$. The controller parameters are set as $K_q = \text{diag}(4, 4, 3)$, $K_{\omega} = \text{diag}(3, 3, 2)$, $f_1 = 3$, $f_2 = 3$, and $f_3 = 3$. In order to reduce the noise in the angular speed measurements, four special rubber dampers are used to join the IMU sensor and the quadrotor body. Therefore, the noise in the angular velocity measurements can be reduced. Besides, in order to show the intuitive physical meaning of the attitude references clearly, the desired attitude references based on unit quaternion representations are obtained by the desired Euler angle references. As illustrated in [23], it can be achieved as follows:

$$\begin{bmatrix} q_1^T(t) \\ q_2^T(t) \\ q_3^T(t) \\ q_4^T(t) \end{bmatrix} = \begin{bmatrix} \cos(\phi^r(t)/2) & \cos(\phi^r(t)/2) & \cos(\phi^r(t)/2) \\ \sin(\phi^r(t)/2) & \cos(\phi^r(t)/2) & \cos(\phi^r(t)/2) \\ \cos(\phi^r(t)/2) & \sin(\phi^r(t)/2) & \cos(\phi^r(t)/2) \\ \cos(\phi^r(t)/2) & \cos(\phi^r(t)/2) & \sin(\phi^r(t)/2) \end{bmatrix} + \begin{bmatrix} \sin(\phi^r(t)/2) & \sin(\phi^r(t)/2) & \sin(\phi^r(t)/2) \\ -\cos(\phi^r(t)/2) & \sin(\phi^r(t)/2) & \sin(\phi^r(t)/2) \\ \sin(\phi^r(t)/2) & \cos(\phi^r(t)/2) & \sin(\phi^r(t)/2) \\ -\sin(\phi^r(t)/2) & \sin(\phi^r(t)/2) & \cos(\phi^r(t)/2) \end{bmatrix} \quad (27)$$

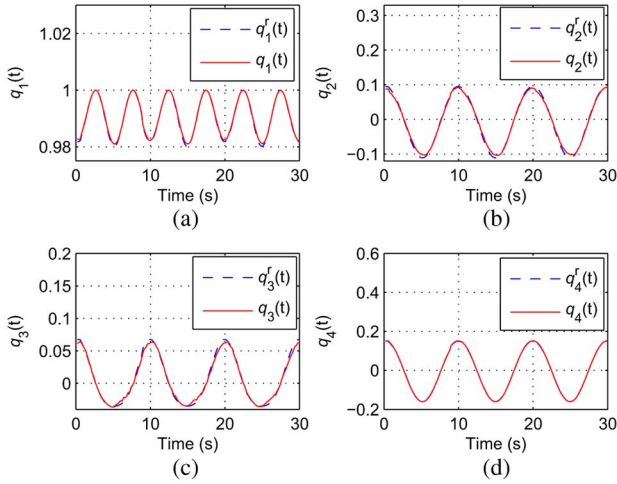


Fig. 4. Attitude responses based on quaternion representations in Case 1. (a) Response of $q_1(t)$. (b) Response of $q_2(t)$. (c) Response of $q_3(t)$. (d) Response of $q_4(t)$.

where $\phi^r(t)$ is the desired roll angle, $\theta^r(t)$ is the desired pitch angle, and $\psi^r(t)$ is the desired yaw angle. The desired rotational speed $\omega_b^r(t)$ is generated by the following equations:

$$\omega_b^r(t) = 2 \begin{bmatrix} -q_2^r(t)\dot{q}_1^r(t) + q_1^r(t)\dot{q}_2^r(t) + q_4^r(t)\dot{q}_3^r(t) - q_3^r(t)\dot{q}_4^r(t) \\ -q_3^r(t)\dot{q}_1^r(t) - q_4^r(t)\dot{q}_2^r(t) + q_1^r(t)\dot{q}_3^r(t) + q_2^r(t)\dot{q}_4^r(t) \\ -q_4^r(t)\dot{q}_1^r(t) + q_3^r(t)\dot{q}_2^r(t) - q_2^r(t)\dot{q}_3^r(t) + q_1^r(t)\dot{q}_4^r(t) \end{bmatrix}.$$

In this section, three experimental tests are implemented to check the tracking performance of the proposed closed-loop system.

A. Case 1

The quadrotor is required to track sine wave references in a coupling condition to check the dynamic tracking performance of the control system. In this case, the sine wave references have 10-s period, and 12° , 18° , and 6° amplitudes for the roll, yaw, and pitch angles, respectively. The helicopter starts from $\phi(0) = 0$, $\theta(0) = 0$, and $\psi(0) = 0$. $q(t)$ is required to follow $q^r(t)$ determined by (27). From the model (1), it can be seen that there exist inter-axis couplings between the attitude angles and the angular velocities. By the proposed robust control method, good tracking performance can be achieved under the influence of various uncertainties as depicted in Fig. 4. In order to make the advantages of the proposed control method clearly, equivalent attitude responses based on the Euler angle representations are shown in Fig. 5 and the tracking errors of the angles in Fig. 6. It can be observed that the tracking errors of the roll, yaw, and pitch angles are about 1.2° , 0.3° , and 0.8° , respectively. In contrast, in [33], a sliding mode attitude controller was proposed for a four-rotor helicopter. The quadrotor was required to track the sine wave references with 20-s period, and 5.73° , 17.19° , and 11.46° amplitudes, for the roll, yaw, and pitch angles, respectively. The experimental tracking errors of the roll, yaw, and pitch angles were approximately 2° , 3° , and 2° , respectively and there were oscillations in the responses of

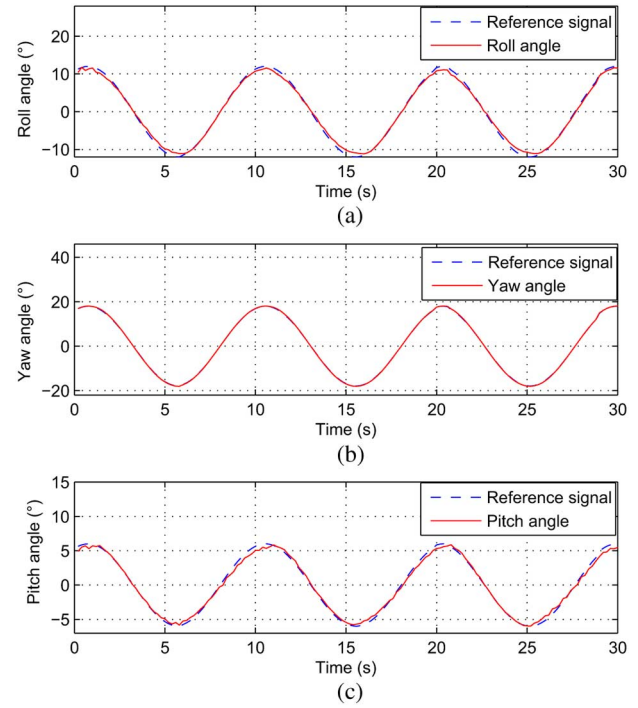


Fig. 5. Attitude responses based on Euler angle representations in Case 1. (a) Response of the roll angle. (b) Response of the yaw angle. (c) Response of the pitch angle.

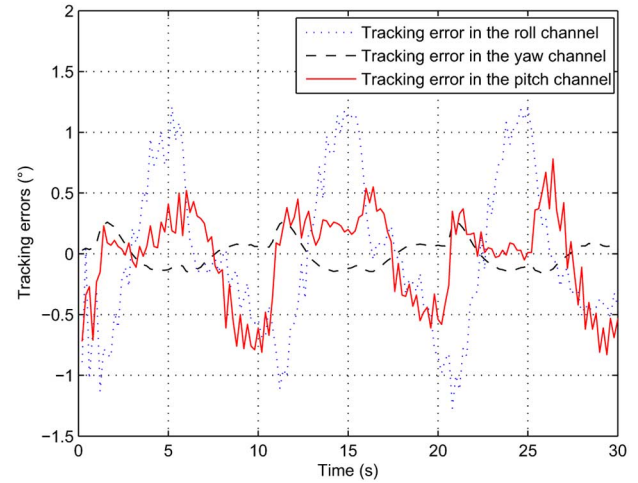


Fig. 6. Tracking errors based on Euler angle representations in Case 1.

the three angles. In [25], a quaternion-based nonlinear feedback controller was developed for the attitude stabilization of a quadrotor mini-helicopter. The yaw angle was required to track the sine wave references with 30-s period, and 18° amplitude, while the roll and pitch angles were needed to stay at 0° . The experimental tracking errors of the roll, yaw, and pitch angles were approximately 4° , 4° , and 4° , respectively. Actually, the effects of uncertainties were not discussed in the stability analysis. In this paper, a continuous attitude controller is applied for uncertain quadrotors to restrain the effects of multiple uncertainties on the closed-loop control system. The steady-state and dynamical tracking performance of the proposed control system can be guaranteed.

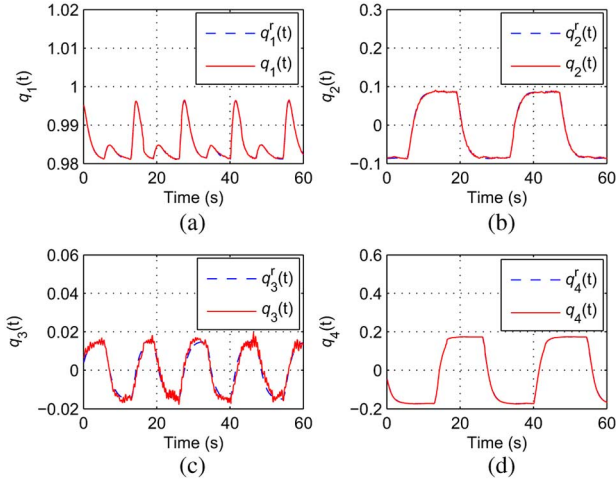


Fig. 7. Attitude responses based on quaternion representations in Case 2. (a) Response of $q_1(t)$. (b) Response of $q_2(t)$. (c) Response of $q_3(t)$. (d) Response of $q_4(t)$.

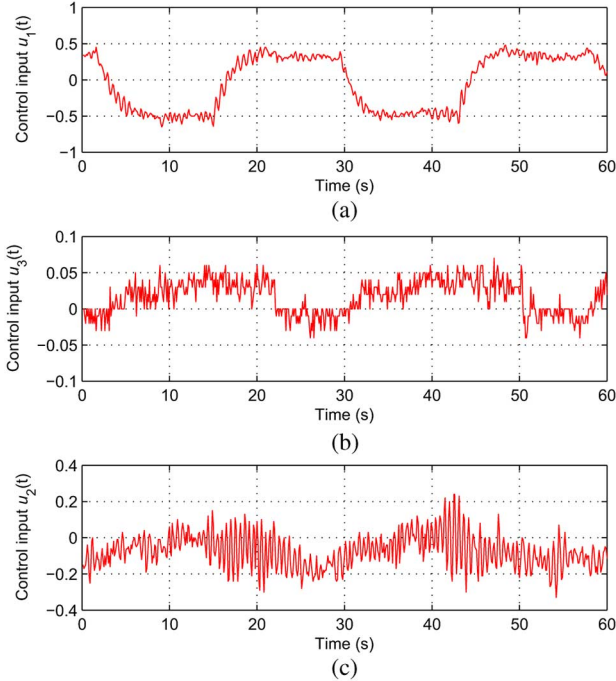


Fig. 8. Control inputs in Case 2. (a) Control input in the roll channel. (b) Control input in the yaw channel. (c) Control input in the pitch channel.

B. Case 2

The steady-state tracking performance of the closed-loop control system will be checked. The vehicle is required to follow the square-like waveform references with 10° and 20° amplitudes for the roll and yaw angles. The attitude responses based on the quaternion representations and the control inputs are shown in Figs. 7 and 8. Equivalent attitude responses based on the Euler angle representations are given in Fig. 9 and the tracking errors of the roll and yaw angles are shown in Fig. 10. From Figs. 9 and 10, it can be seen that when the roll and yaw angles are approximately 10° and 20° , the steady-state tracking errors of the roll, yaw, and pitch angles are only 0.3° , 0.1° ,

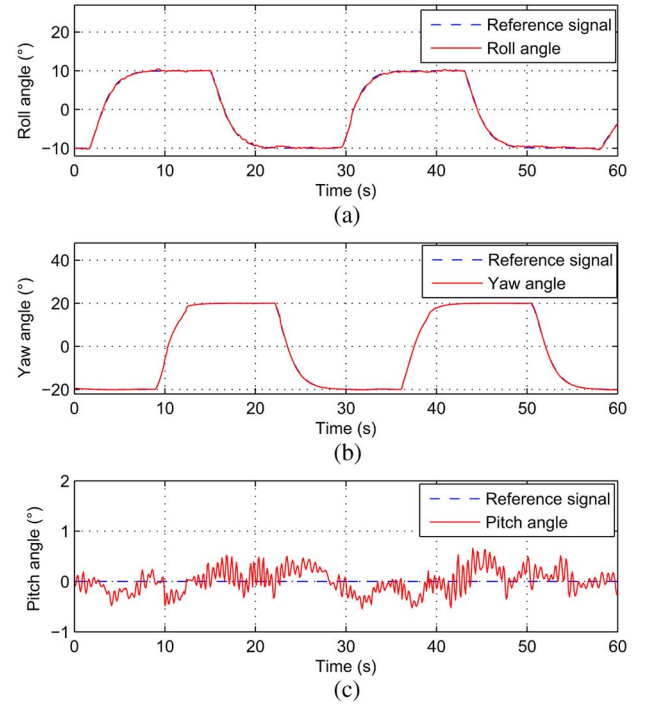


Fig. 9. Attitude responses based on Euler angle representations in Case 2. (a) Response of the roll angle. (b) Response of the yaw angle. (c) Response of the pitch angle.

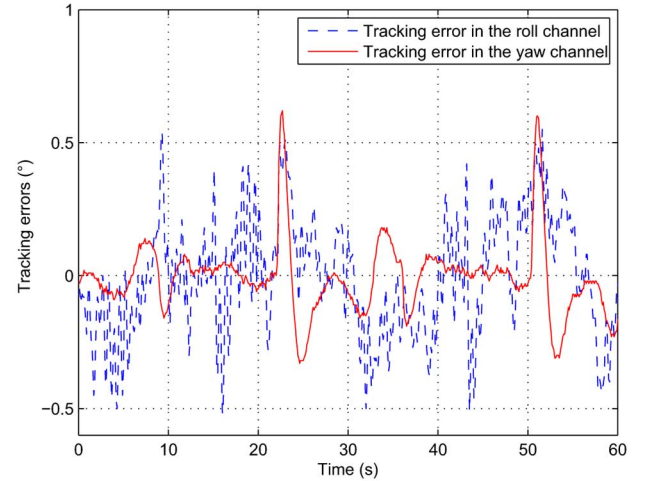


Fig. 10. Tracking errors based on Euler angle representations in Case 2.

and 0.3° , respectively, from 12 to 15 s. The closed-loop control system achieves good steady-state tracking performance in coupling condition. From Fig. 7, one may feel that $q_3(t)$ cannot track $q_3^r(t)$ as well as other quaternions, but the absolute tracking error of $q_3(t)$ is similar to $q_2(t)$ and $q_4(t)$.

C. Case 3

The vehicle is required to track sine wave references in a coupling condition under the effect of wind gusts as depicted in Fig. 3. The electrical fan is switched ON at approximate 0 s and switched OFF at approximate 5 s. Then, it is switched ON again at approximate 10 s and switched OFF at approximate

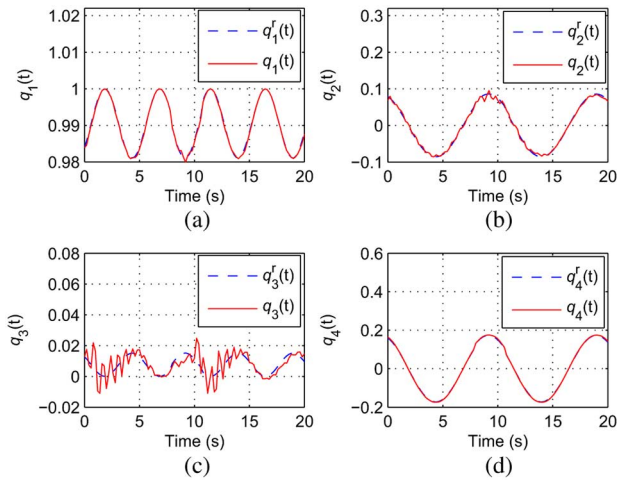


Fig. 11. Attitude responses based on quaternion representations in Case 3. (a) Response of $q_1(t)$. (b) Response of $q_2(t)$. (c) Response of $q_3(t)$. (d) Response of $q_4(t)$.

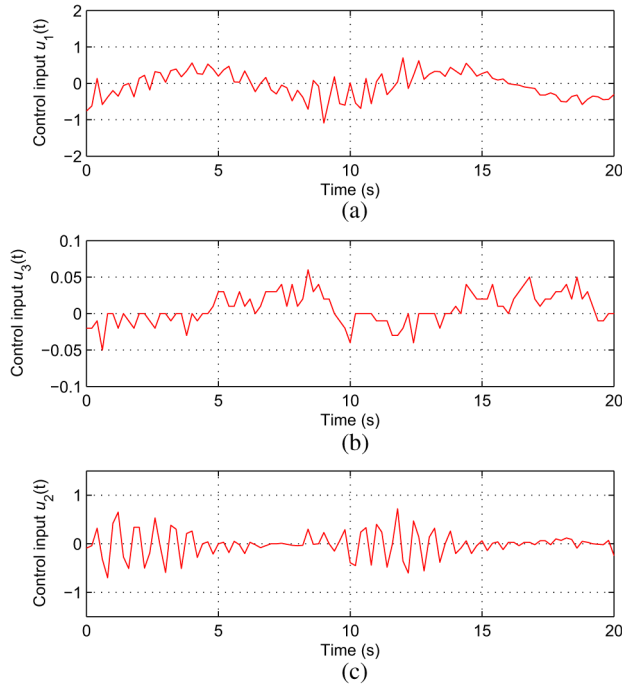


Fig. 12. Control inputs in Case 3. (a) Control input in the roll channel. (b) Control input in the yaw channel. (c) Control input in the pitch channel.

15 s. An anemometer with the effective resolution of 0.1 m/s is used to measure the wind velocity. When the fan is switched ON, it can produce the wind gusts with 4.2 m/s velocity. The roll and yaw angles are required to track the sine wave references with 10-s period, and 10° and 20° amplitudes, respectively. The pitch is required to stay at 0° . The attitude responses based on the quaternion representations and corresponding control inputs are depicted in Figs. 11 and 12. Equivalent attitude responses based on the Euler angle representations are shown in Fig. 13 and tracking errors of the roll and yaw angles are depicted in Fig. 14. One can see that when the closed-loop control system encounters the effects of wind gusts, tracking errors are 2.2° ,

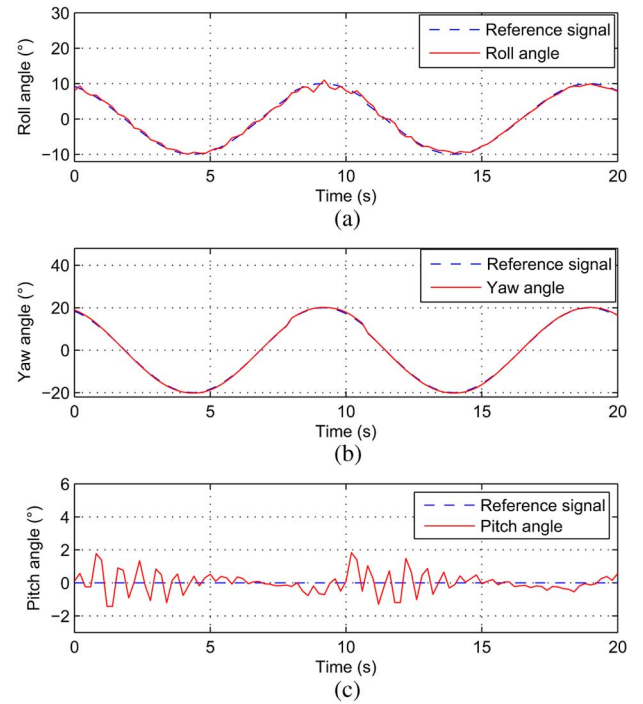


Fig. 13. Attitude responses based on Euler angle representations in Case 3. (a) Response of the roll angle. (b) Response of the yaw angle. (c) Response of the pitch angle.

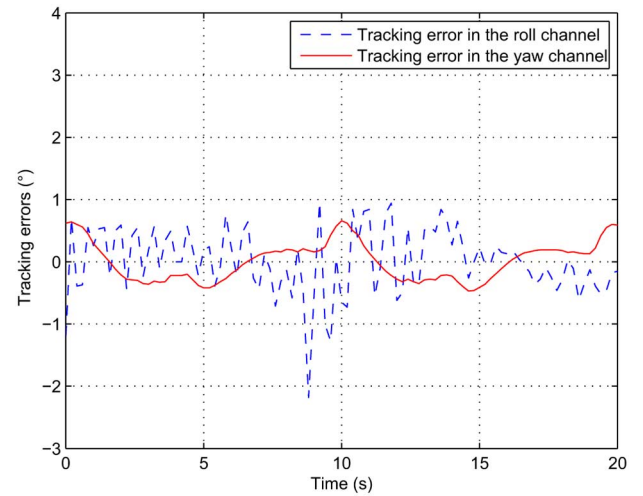


Fig. 14. Tracking errors based on Euler angle representations in Case 3.

0.7° , and 2.0° , for the roll, yaw, and pitch angles in the sine wave reference tracking mission. In contrast, previous studies focused on wind gust rejection studies for quadrotors mainly in hovering missions by simulations (see, [34], [35]). In [36], experimental results with only large reference tracking for the yaw channel were given to show the effectiveness of the robust attitude control method for an unmanned helicopter mounted on a spherical joint in hovering missions under the effects of 4.9 m/s wind gusts. In this paper, the quadrotor achieves robust attitude control in large-angle reference tracking missions and coupling condition under wind gusts.

V. CONCLUSION

A robust controller design method is proposed for the attitude control problem of the quadrotors. It consists of a nominal controller and a robust compensator. The nominal controller determines the desired poles of the nominal closed-loop control system and the robust compensator is introduced to attenuate the effects of the input equivalent disturbances. The attitude tracking errors and the rotational velocity tracking errors are proven to be bounded within a specified neighbourhood of the origin. Experimental results on the quadrotor demonstrate the ability of the proposed control method to deal with uncertainties.

Further work can be done on applying the proposed quaternion-based robust control approach on the visual servo control system of quadrotors. It is challenging for the quadrotors to implement the aggressive acrobatic missions.

REFERENCES

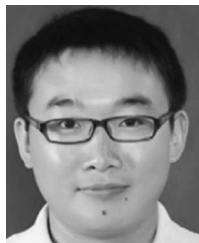
- [1] M. O. Efe, "Neural network assisted computationally simple control of a quadrotor UAV," *IEEE Trans. Ind. Informat.*, vol. 7, no. 2, pp. 354–361, May 2011.
- [2] M. L. Civita and G. Papageorgiou, "Design and flight testing of an H_∞ controller for a robotic helicopter," *J. Guid. Control Dyn.*, vol. 29, no. 2, pp. 485–494, Mar./Apr. 2006.
- [3] G. Cai, B. Chen, K. Peng, M. Dong, and T. H. Lee, "Modeling and control of the yaw channel of a UAV helicopter," *IEEE Trans. Ind. Electron.*, vol. 55, no. 9, pp. 2426–2434, Sep. 2008.
- [4] Y. Tang, H. Gao, J. Kurths, and J. Fang, "Evolutionary pinning control and its application in UAV coordination," *IEEE Trans. Ind. Informat.*, vol. 8, no. 4, pp. 828–838, Nov. 2012.
- [5] Y. Du, J. Fang, and C. Miao, "Frequency domain system identification of an unmanned helicopter based on adaptive genetic algorithm," *IEEE Trans. Ind. Electron.*, vol. 61, no. 2, pp. 870–881, Feb. 2014.
- [6] H. Liu, X. Wang, and Y. Zhong, "Robust position control of a lab helicopter under wind disturbances," *IET Control Theory Appl.*, vol. 8, no. 15, pp. 1555–1565, Oct. 2014.
- [7] F. Lin, X. Dong, B. M. Chen, K. Y. Lum, and T. H. Lee, "A robust real-time embedded vision system on an unmanned rotorcraft for ground target following," *IEEE Trans. Ind. Electron.*, vol. 59, no. 2, pp. 1038–1049, Feb. 2012.
- [8] H. Liu, J. Xi, and Y. Zhong, "Robust hierarchical control of a laboratory Helicopter," *J. Franklin Inst.*, vol. 351, no. 1, pp. 259–276, Jan. 2014.
- [9] L. Marconi and R. Naldi, "Robust full degree-of-freedom tracking control of a helicopter," *Automatica*, vol. 43, no. 11, pp. 1909–1920, Nov. 2007.
- [10] E. Prempain and I. Postlethwaite, "Static H_∞ loop shaping control of a fly-by-wire helicopter," *Automatica*, vol. 41, no. 9, pp. 1517–1528, Sep. 2005.
- [11] F. Kendoul, D. Lara, I. Fantoni-Coichot, and R. Lozano, "Real-time nonlinear embedded control for an autonomous quadrotor helicopter," *J. Guid. Control Dyn.*, vol. 30, no. 4, pp. 1049–1061, Jul./Aug. 2007.
- [12] G. M. Hoffmann, H. Huang, S. L. Waslander, and C. J. Tomlin, "Quadrotor helicopter flight dynamics and control: Theory and experiment," in *Proc. AIAA Guid. Navig. Control Conf. Exhibit.*, Hilton Head, SC, USA, Aug. 2007, pp. 1–20.
- [13] P. Pounds, R. Mahony, and P. Corke, "Modelling and control of a large quadrotor robot," *Control Eng. Pract.*, vol. 18, no. 7, pp. 691–699, Jul. 2010.
- [14] E. Altug, J. P. Ostrowski, and C. J. Taylor, "Control of a quadrotor helicopter using dual camera visual feedback," *Int. J. Robot. Res.*, vol. 24, no. 5, pp. 329–341, May 2005.
- [15] P. Castillo, A. Dzul, and R. Lozano, "Real-time stabilization and tracking of a four-rotor mini rotorcraft," *IEEE Trans. Control Syst. Technol.*, vol. 12, no. 4, pp. 510–516, Jul. 2004.
- [16] S. Bertrand, N. Guenard, T. Hamel, H. Piet-Lahanier, and L. Eck, "A hierarchical controller for miniature VTOL UAVs: Design and stability analysis using singular perturbation theory," *Control Eng. Pract.*, vol. 19, no. 10, pp. 1099–1108, Jul. 2011.
- [17] C. Aguilar-Ibanez, H. Sira-Ramirez, M. S. Suarez-Castanon, E. Martinez-Navarro, and M. A. Moreno-Armendariz, "The trajectory tracking problem for an unmanned four-rotor system: Flatness-based approach," *Int. J. Control*, vol. 85, no. 1, pp. 69–77, Jan. 2012.
- [18] L. Besnard, Y. B. Shtessel, and B. Landrum, "Quadrotor vehicle control via sliding mode controller driven by sliding mode disturbance observer," *J. Franklin Inst.*, vol. 349, no. 2, pp. 658–684, Mar. 2012.
- [19] L. Luque-Vega, B. Castillo-Toledo, and A. G. Loukianov, "Robust block second order sliding mode control for a quadrotor," *J. Franklin Inst.*, vol. 349, no. 2, pp. 719–739, Mar. 2012.
- [20] Z. T. Dydek, A. M. Annaswamy, and E. Lavretsky, "Adaptive control of quadrotor UAVs: A design trade study with flight evaluations," *IEEE Trans. Control Syst. Technol.*, vol. 21, no. 4, pp. 1400–1406, Jul. 2013.
- [21] Z. Zuo, "Trajectory tracking control design with command-filtered compensation for a quadrotor," *IET Control Theory Appl.*, vol. 4, no. 11, pp. 2343–2355, Nov. 2010.
- [22] T. Ryan and H. J. Kim, "LMI-based gain synthesis for simple robust quadrotor control," *IEEE Trans. Autom. Sci. Eng.*, vol. 10, no. 4, pp. 1173–1178, Oct. 2013.
- [23] A. Tayebi and S. McGilvray, "Attitude stabilization of a VTOL quadrotor aircraft," *IEEE Trans. Control Syst. Technol.*, vol. 14, no. 3, pp. 562–571, May 2006.
- [24] E. N. Johnson and S. K. Kannan, "Adaptive trajectory control for autonomous helicopters," *J. Guid. Control Dyn.*, vol. 28, no. 2, pp. 524–538, May/Jun. 2005.
- [25] J. F. Guerrero-Castellanos, N. Marchand, A. Hably, S. Lasecq, and J. Delamare, "Bounded attitude control of rigid bodies: Real-time experimentation to a quadrotor mini-helicopter," *Control Eng. Pract.*, vol. 19, no. 8, pp. 790–797, Aug. 2011.
- [26] R. Zhang, Q. Quan, and K. Y. Cai, "Attitude control of a quadrotor aircraft subject to a class of time-varying disturbances," *IET Control Theory Appl.*, vol. 5, no. 9, pp. 1140–1146, Jun. 2011.
- [27] A. Isidori, L. Marconi, and A. Serrani, "Robust nonlinear motion control of a helicopter," *IEEE Trans. Automat. Control*, vol. 48, no. 3, pp. 413–426, Mar. 2003.
- [28] G. Hu, N. Gans, N. Fitz-Coy, and W. Dixon, "Adaptive homography-based visual servo tracking control via a quaternion formulation," *IEEE Trans. Control Syst. Technol.*, vol. 18, no. 1, pp. 128–135, Jan. 2010.
- [29] H. Liu, Y. Q. Bai, G. Lu, and Y. S. Zhong, "Robust Attitude Control of Uncertain Quadrotors," *IET Control Theory Appl.*, vol. 7, no. 11, pp. 1583–1589, Jul. 2013.
- [30] Y. Zhong, "Robust output tracking control of SISO plants with multiple operating points and with parametric and unstructured uncertainties," *Int. J. Control*, vol. 75, no. 4, pp. 219–241, Apr. 2002.
- [31] H. Liu, G. Lu, and Y. Zhong, "Robust LQR attitude control of a 3-DOF lab helicopter for aggressive maneuvers," *IEEE Trans. Ind. Electron.*, vol. 60, no. 10, pp. 4627–4636, Oct. 2013.
- [32] B. Zheng and Y. Zhong, "Robust attitude regulation of a 3-DOF helicopter benchmark: Theory and experiments," *IEEE Trans. Ind. Electron.*, vol. 58, no. 2, pp. 660–670, Feb. 2011.
- [33] L. Derafa, A. Benallegue, and L. Fridman, "Super twisting control algorithm for the attitude tracking of a four rotors UAV," *J. Franklin Inst.*, vol. 349, no. 2, pp. 685–699, Mar. 2012.
- [34] K. A. Danapalasingam, J. J. Leth, A. L. C. Harbo, and M. Bisgaard, "Robust helicopter stabilization in the face of wind disturbance," in *Proc. IEEE Conf. Decis. Control*, Atlanta, GA, USA, 2010, pp. 3832–3837.
- [35] B. Ahmed and F. Kendoul, "Flight control of a small helicopter in unknown wind conditions," in *Proc. IEEE Conf. Decis. Control*, Atlanta, GA, USA, 2010, pp. 3536–3541.
- [36] F. Leonard, A. Martini, and G. Abba, "Robust nonlinear controls of model-scale helicopters under lateral and vertical wind gusts," *IEEE Trans. Control Syst. Technol.*, vol. 20, no. 1, pp. 154–163, Jan. 2012.



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